Dual-ligand fluorination

01 / Experiment

Use the two-page bench sheet at the robot. Keep the protocol and its generated CSVs together; `dosing.csv` gives well identities and `deck_loads.csv` gives stage-specific source fills.

Reaction and design

Pyridine-2-sulfonyl fluoride (PyFluor) to 2-fluoropyridine. Per vial: 25 µmol SM, 1.5 µmol each ligand, 3 µmol Pd(COD)(DQ), 2.5 µmol mesitylene; 250 µL toluene, 95 °C, 500 rpm, 4 h under air. No base or external fluoride.

Round one: **48 distinct pairs, all 30 available ligands**; L07 is unavailable. Round two: **22 new pairs + L17 tBuBrettPhos at 12 mol% total ligand**. No repeats. A better pair yield establishes improvement against this reference; it does not establish reproducibility or ligand synergy.

Before the clock starts

- Fit **p300 single GEN2 left; p300 eight-channel GEN2 right**. Reaction vials have 13 mm pitch: use the single only. The multi serves the standard 96-well LC preparation plate.
- Check the supplied block on the flat adapter: secure mounting, latch closure, cap clearance within **45 mm**, offsets and thermal transfer. Pipette **1.5 mm above the bottom** by default; check actual immersion. Caps are manual; no stir bars or gripper.
- Confirm the existing **1.7-min full LC cycle**, method ID, injection volume, UV channel/wavelength and authentic-standard RT/LOQ settings. Requested 210 nm is provisional; the old example wavelengths are not our method. Record local evidence before release.
- Prepare stocks, caps, tips, 71 fresh reaction vials, 71 sample HPLC vials plus standards, and two LC preparation plates. App-simulate the released Python. Open setup details.

Slot	Load
1 / 4	Ligand source racks; exact CSV addresses, L07 absent
2	A1 SM/DCM; A2 Pd; A3 toluene; A8 mesitylene. Quench A4/A6/A9; diluent A5/A7/A10/A11/A12
3 / 10	Vial block for pipetting / heater-shaker with universal flat adapter
6	Standard 96-well LC preparation plate; keep seated during sampling
5 / 8 / 9	Single tips / multi tips / multi tips. Keep slots 7 and 11 empty

Dose and react

1. Robot doses **50 µL 500 mM SM/DCM** per vial. At the pause, move the block off deck and evaporate completely by HTEL's ventilated procedure. Cool; return to slot 3.
2. Add **25 µL mesitylene +37.5 µL ligand A +37.5 µL ligand B +75 µL Pd +75 µL toluene =250 µL**. Stocks: mesitylene 100 mM; ligands/Pd 40 mM. Reference: use 75 µL L17 instead of the two ligand portions.
3. Hand-tighten all caps. Move/latch at the local pauses; run **95 °C, 500 rpm, 4 h**. Every block move, cap change and tip-rack replacement is manual.
4. During hold one, run the **14 calibration injections**. Hand-dose **23 labelled fresh vials with SM/DCM only** in an ordinary off-deck rack; evaporate fully and cap the dry SM. Add mesitylene/catalysts only after round two is selected. Reuse the same reaction block.

Timing: 717.7 min planned against 720 min, requiring 35 min LC/handling overlap. This remains unmeasured, with about two minutes spare. Record actual heating, cooling, cap handling and LC cycle times.

Workup and LC sampling

1. **Cool sealed vials off the heater.** Open for addition; add **500 µL 9:1 ACN/water +0.15% formic acid**. Nominal extract: 750 µL, 0.1% acid, 1.25 mL headspace. Quench and diluent contain no IS.
2. **Hand-recap every vial.** Shake **1500 rpm, 2 min**, heater off; return to slot 3. The block must have cooled for workup.
3. **Sample eight at a time.** Gently remix the indicated capped vials; let bubbles settle. Open only that batch. Other vials stay capped.
4. **Robot dilution:** fresh tip per sample; two slow same-well prewet cycles, without blowing air into the extract. Aspirate 21 µL at 1.5 mm, retain 1 µL reverse surplus, add 2 µL air gap and dispense nominal **20 µL into 980 µL LC diluent**, noncontact. Discard the tip; never revisit the source after destination contact.
5. **At the pause:** recap sampled reaction vials. Keep the preparation plate seated. With a fresh manual tip per completed LC well, mix **500 µL five times**, avoiding air; transfer **900 µL** to the labelled HPLC vial. Cap and invert briefly.
6. **Start that LC batch before resuming the robot.** Never transfer unfinished wells. R1: six batches of eight. R2: 8/8/7; preparation well **H3 is diluent only and must not be injected**. Reaction A1-F8 and LC A1-H6 are different maps.

Final sample: 1.00 mL; maximum SM/product **666.7 µM**; mesitylene **66.7 µM**; formic acid **0.1%**; toluene **0.667%**. Set injection volume from the actual calibrated method.

Pipetting and transfer faults

Condition fresh dosing tips twice in their own stock. Use slow aspiration, short settling delays, a small air gap and reverse surplus for the organic stocks. Keep active sources covered between stages and bulk reserves capped. Follow CSV fills and dead volumes; never return working liquid to bulk stock.

Reuse a tip only for consecutive **same-stock, noncontact** dosing above the destination top. Different ligands and all analytical samples use fresh tips. Avoid deep immersion, walls and bottoms. Repeated aspiration does not dissolve an insoluble stock and may introduce bubbles.

Pause for persistent layers, solids at the aspiration point, bubbles, air uptake, dripping or tip contact. Do not sample an arbitrary layer. Do not filter, centrifuge or decant without recording its effect on analyte/IS recovery. High rpm alone does not establish quench completeness or extraction recovery; no separate homogeneity study is planned.

For an uncertain LC dilution, remix the original extract and prepare a fresh documented dilution. Do not guess a top-up. Retain rejected measurements with a reason and accept one result per sample; do not select the highest yield.

Changeover and finish

After all **48 R1 results** pass analytical QC, the local pipeline selects **22 new pairs + L17 reference**. Archive R1 vials; fit the 23 dry predosed vials into the same block and verify orientation. Refill sources using R2 deck loads. **No SM redosing; mesitylene is added once after dry-down.** Repeat reaction and workup.

Stop the clock after all 23 final injections and quantitative analysis. Archive source exports, maps, assay/stock records, method/channel, calibration, QC, times, scoring/cache records, final CSVs, matching protocols and robot JSON logs. Keep experimental data private pending team agreement.

02 / Stocks and calibration

Use `stock_preparation.csv` for masses/reserves; load only the stage's `deck_loads.csv` quantities. **Bulk volume is not deck fill.** Correct masses for COA assay. Prepare ligands in the glovebox.

Reaction stocks

Stock	Bulk preparation at 100% assay	Per reaction
SM, MW 161.16	**10.00 mL, 500 mM in DCM:** 805.8 mg; make to volume in a capped 20 mL glass vial	50 µL; evaporate fully
Each ligand	**40 mM in toluene**; masses and reserves in CSV	37.5 µL each
Pd(COD)(DQ), MW 378.8	**40 mM in toluene**; current plan 9.53 mL, 144.399 mg	75 µL
Mesitylene, MW 120.19	**5.00 mL, 100 mM in toluene:** 60.095 mg	25 µL, after dry-down
Toluene	Current reserve plan 15.57 mL	75 µL

Capped glass bulk reserves; ordinary plastic working racks, maximum 1.4 mL/tube; common liquids in the 12-channel reservoir. CSVs include 50% working margin, dead volume and conservative R2 reserves. Refill stage sources; never return working liquid to bulk.

Workup solvents - no internal standard

Solution	Prepare	Use
Quench	**100 mL, 9:1 ACN/water +0.15% v/v formic acid:** 150 µL acid, make to volume with premixed solvent	500 µL; recap, 1500 rpm, 2 min
LC diluent	**125 mL, 1:1 water/ACN +0.1% v/v formic acid:** 125 µL acid, make to volume with premixed solvent	980 µL +20 µL extract

Analytical stocks - matched matrix

Matrix, 100 mL: 0.6667 mL toluene +50.1492 mL ACN +0.1000 mL formic acid; water to volume. Approximate diluted-extract matrix; ignores volume contraction. Prepare all analytical stocks in it.

Separate 10 mM analyte masters, 10.00 mL: SM 16.116 mg; product 9.709 mg. No IS. For 1 mM working stocks, dilute 100 µL master to 1.00 mL. Make check masters by **independent weighing**.

Analytical mesitylene, 5 mM, 20.00 mL: 12.019 mg in matched matrix. Do not use the reaction mesitylene/toluene stock.

Calibration during hold one - 14 injections

Prepare seven **1.00 mL vials per analyte**. Every vial, including blank/check, receives **13.333 µL of 5 mM mesitylene** (66.667 µM final IS). Add the quantities below; cap, mix and use `calibration_template.csv` IDs.

Analyte, µM	Role	Stock, mM	Analyte, µL	Matrix, µL
0	Fit blank	None	0	986.667
10	Fit	1	10	976.667
50	Fit	1	50	936.667
150	Fit	10	15	971.667
350	Fit	10	35	951.667
800	Fit	10	80	906.667
666.667	Independent check	10, independent	66.667	920.000

Fit blank +five levels; exclude the check. Require **R^2 >=0.995**, **check error <=15%**, separation and measured LOQs on the actual method. The lowest standard is not automatically the LOQ. Record method, wavelength and injection volume.

Maximum sample: 666.667 µM analyte; 64.7 µg/mL product or 107.4 µg/mL SM. Calibration must establish the detector range. LC import and pilot replay covers reviewed blank absences and optional mixed-standard practice. Retain failed checks/rejected measurements.

03 / Local pipeline

The **initial 48 pairs are ready** in `campaigns/hackathon2026/round1_design.csv`. Use the local workstation. Chromeleon exports UV areas; Python applies our standards/calibration and calculates yields. Staff use the bench sheet.

Set up before the clock

Use Python 3.12. Install with `uv sync --python 3.12 --extra sources` and prefix commands with `uv run`, or activate a virtual environment and run `python -m pip install -e '.[sources]'`.

```
mkdir -p local_data
cp -n hte_inputs/campaign.json local_data/campaign.json
cp -n hte_inputs/calibration_template.csv local_data/calibration-plan.csv
cp -n anthropic.key.example anthropic.key
chmod 600 anthropic.key
```

Keep an existing key. Replace the placeholder in `anthropic.key` with the raw Claude key. Keep it local; never include it in robot files, Git or email.

In `local_data/campaign.json`, enter method ID, full LC cycle, UV channel/wavelength, standard RT windows and measured SM/product LOQs. Record applicable `validation` evidence; HTEL checks. Requested 210 nm is provisional. Review visible transfer faults; no separate homogeneity study is required.

```
python -m hte.cli --config local_data/campaign.json check
python -m hte.cli --config local_data/campaign.json export \
  --design campaigns/hackathon2026/round1_design.csv \
  --output output/r1-dose --live --stage dose_and_react
```

App-import/simulate `output/r1-dose/dose_and_react.py`; use matching CSVs and start locally. Before setup is complete, omit `--live`; keep the simulation guard. Dosing can precede calibration once other applicable checks and the method are recorded.

During hold one

Run/import **14 calibration injections** using the LC instructions. Staff prepare 23 dry SM-only vials. After calibration passes, set `lc_calibration_validated=true`; export workup:

```
python -m hte.cli --config local_data/campaign.json export \
  --design campaigns/hackathon2026/round1_design.csv \
  --output output/r1-workup --live --stage workup
```

After all 48 R1 results

Complete `lc_batchNN_injections.csv` maps with actual method/sequence identities. Each export needs its own map, including reset injection numbers. Keep both globs in matching batch order; exclude unrelated files.

```
python scripts/hte-lc-results.py next \
  --dosing output/r1-dose/dosing.csv \
  --exports local_data/r1-batch*.csv \
  --injections local_data/r1-map*.csv \
  --output output/r1-decision
```

Calibration defaults to `local_data/calibration.csv`; override with `--calibration` if needed. The command checks all 48 samples/calibration/QC, then scores the **387 untested pairs with Claude** and outputs **22 new pairs + L17 reference**. Both prompts include the same single-ligand SI yields. Round two uses no embedding/GoLLuM selector.

The printed decision directory holds `next_design.csv`, `next_dosing.csv`, analysis, scores and `protocols/`. Use released Python and matching R2 deck loads. If release checks remain open, complete them and export from `next_design.csv` with `--history output/r1-dose/dosing.csv`, a fresh output folder and `--live`. Replace R1 vials with the 23 dry predosed vials; **do not redose SM**.

After all 23 R2 injections, run the helper with `final`, R2 `dosing.csv`, three matching export/map batches and a new output folder. This calculates final results without Claude. Compare pair yields with the measured L17 reference and retain run times.

Practice and recovery

For analytical challenges and pilot recipes, run `python -m hte.cli practice-analytics --output output/pilot`. Real pilot replay and blank reviews. The full `rehearse-loop` includes this plus mocked feedback. No paid calls. Use fresh import folders; review uncertain scoring attempts before retrying.

159 tests passed; protocols simulated; initial design reproducible. Bench results/calibration and feedback latency remain unmeasured. Validation, ML strategy, PDF updates.